

POCKET PRODUCER: SESSION-CONDITIONED RETRIEVAL AS USER-CONTROLLED PROPOSAL

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ABSTRACT

Music continuation is not a similarity problem: the loop that best matches a session is not necessarily the one that helps a musician move it forward. Pocket Producer operationalizes this distinction inside the Audiotool DAW: it fingerprints the live session via the Nexus SDK (tempo, CLAP embeddings of session audio) and ranks Freesound Loop Dataset candidates with a five-signal fusion trained on pack co-membership weak labels—the only supervision available cold-start. Offline evaluation shows the apparent ranking gain changes substantially with weak-label construction, negative regime, and embedding representation. Since the proxy does not encode the creative objective, every recommendation is an inspectable, reversible *proposal*—previewed with per-signal evidence, inserted in a single undoable transaction, logged for future personalization—rather than an autonomous creative decision.

1. INTRODUCTION

When a musician returns to an unfinished loop-based project, the question is not “what sounds similar?” but “what would be useful as the next step?” Loop compatibility [1–4], stem coherence [5–7], session-conditioned generation [8], and reversible co-creation [9] all address parts of this problem. We define **continuation utility** as the property that a candidate helps the musician make progress in the current session; unlike similarity it is asymmetric and context-dependent.

We contribute: (1) a problem formulation that separates continuation utility from similarity and compatibility; (2) empirical evidence that ranking conclusions drawn from a weak proxy shift with label construction, negative regime, and embedding representation; (3) a deployed system that turns this finding into interaction design: proposal-based retrieval with per-signal evidence, preview, reversible insertion, and feedback logging.

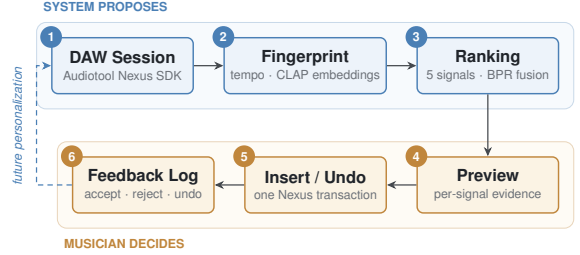


Figure 1. Pocket Producer pipeline: the system proposes (steps 1–3); the musician decides (steps 4–6).

2. SYSTEM

2.1 Session Fingerprinting

The live Audiotool session (Figure 1) is read via the Nexus SDK, which exposes the project’s devices, regions, and referenced audio samples in real time. The system extracts the project tempo from the document metadata and collects up to eight unique audio samples referenced by regions in the arrangement. Each sample is embedded with CLAP [10]; the resulting vectors form the session’s timbral fingerprint. When no session audio is available (empty project or no sample-backed regions), the system falls back to non-audio signals only.

2.2 Multi-Signal Ranking

Five signals score each FSLD [11] candidate against the session fingerprint: (i) mean-pooled and (ii) max-pooled CLAP cosine similarity to session samples, (iii) tempo proximity, (iv) key compatibility, and (v) tag Jaccard overlap between candidate tags and tags from the session’s audio fragments. A logistic fusion trained with BPR loss [12] on Freesound pack co-membership weak labels (Section 3) combines these into a single score. A hand-tuned *rules* baseline using only tempo, key, and tag signals serves as the default ranker; the fusion is selectable and falls back to rules when no session audio is available.

2.3 Proposal Interface

The title’s “user-controlled proposal” is a deliberate design choice, not a fallback from poor accuracy. Pack co-membership is a proxy for timbral compatibility, not continuation utility; even a perfect ranker under this label would not capture arrangement context, creative intent, or genre conventions. The design question is not whether



the ranker can optimize the proxy, but whether the proxy should hold final creative authority; the interface answers by keeping the musician in the loop at every step.

Each candidate is surfaced for in-context preview with per-signal evidence—cosine score, tempo delta, key match, and overlapping tags—so the musician can assess *why* the system ranks it highly and judge whether those reasons apply to their current goal. An accepted candidate is inserted via a single Nexus transaction; *undo* reverts the insertion and retracts the candidate’s tags from the ranking context, so subsequent recommendations are not biased by the undone choice. Acceptance, rejection, and undo events are logged with session context for future personalization (not yet evaluated).

2.4 Deployed Model

The deployed ranker follows the same five-signal recipe but is a separate production model with four stated deltas: MS-CLAP embedding space (offline evaluation uses LAION-CLAP Music); source-weighted BPR (hard_similar $\times$ 2.0, mean over 5 seeds); key constant zero online (Audiotool documents lack key-signature metadata); tag context drawn from session fragments rather than pack siblings.

3. WEAK-LABEL FUSION

From the FSLD [11] we kept only sounds with a true Freesound pack (1,797 items, 605 packs), discarding items grouped only by uploader. Each eligible pack with at least three items yields one leave-one-out query per item, with same-pack siblings as positives and cross-pack items as negatives in three difficulty tiers (*easy*: tempo/key mismatch; *hard_tempo_key*: compatible tempo and key; *hard_similar*: highest cosine to the positive from another pack). A 121-source held-out split (seed 20260810) was committed to version control together with the evaluation protocol *before* any evaluation ran. Development uses only the 484 train sources; the 171 with ≥ 3 items form queries (981 examples), the rest contribute negatives only. LAION-CLAP Music [13] (512-dim) provides the embedding backbone.

On the train side (mean over 5 source-grouped splits), frozen cosine reaches 0.929 overall and 0.814 on hard_similar; five-signal fusion reaches 0.925 and 0.813. Under pack-only labels the fusion advantage is gone: -0.03 pp on hard_similar, and fusion trails cosine on hard_tempo_key by 2.1 pp on average (negative in all 5 splits). These development-set numbers are descriptive; inference is reserved for the held-out analysis (Section 4).

4. OBSERVATIONS

The weak label measures pack co-membership, not continuation utility; the observations below chart where the proxy breaks down—and the feedback log of Section 2.3 is designed to eventually ground ranking in in-session

evidence instead. Four lines of evidence jointly motivate treating the fusion as a proposal rather than a decision. **Label sensitivity.** An earlier corpus treated same-uploader items as co-members (26% of examples, 949 sources); fusion appeared $+4.0$ pp over frozen cosine on hard_similar. Restricting to true pack co-membership erases the gain (Section 3). **Regime dependence.** Leave-one-signal-out ablation on the dev set shows tempo helps on hard_similar (-0.96 pp when removed) but hurts on hard_tempo_key ($+0.68$ pp); signals switch sign across negative regimes, so a fused score averaging over them cannot be uniformly trusted. **Held-out uncertainty.** On a pre-committed 121-source split, the fusion–cosine advantage on hard_similar is $+1.5$ pp (bootstrap 95% CI $[-1.5, +4.7]$, crossing zero) under LAION-CLAP; under MS-CLAP with source-weighted BPR it is $+9.2$ pp (CI $[+4.3, +14.3]$)—a 7.7 pp swing evidencing representation sensitivity. **Pilot divergence.** Nine raters on 20 blind A/B pairs (Fleiss’ $\kappa = 0.294$) show a descriptive divergence between weak-label signal rankings and human consensus.

5. CONCLUSION

Pocket Producer operationalizes session-conditioned retrieval as a user-controlled proposal in a production DAW. It reads the live session via the Nexus SDK, ranks candidates with a multi-signal fusion, and surfaces per-signal evidence for preview; every insertion is reversible and every interaction is logged. Since offline evaluation shows the fusion advantage is sensitive to label construction, embedding space, and negative regime, the default ranker is a transparent rules baseline; the fusion serves as a selectable alternative. Evaluating whether preview and undo improve creative outcomes, and personalization from logged feedback, is future work.

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